2.5.3 Newmark Methods

As a special case of a partitioned Runge-Kutta method, consider the Newmark family of methods [280] defined for two parameters σ and η by the formulas

$$\boldsymbol{P} = \boldsymbol{p} - h(1-\sigma)\nabla U(\boldsymbol{q}) - h\sigma\nabla U(\boldsymbol{Q}),$$
$$\boldsymbol{Q} = \boldsymbol{q} + h\boldsymbol{M}^{-1}\boldsymbol{p} - h^2\left(\frac{1}{2}-\eta\right)\nabla U(\boldsymbol{q}) - h^2\eta\nabla U(\boldsymbol{Q}).$$

This is an exact transcription of the formulas in Newmark's original paper, only substituting $\boldsymbol{M}^{-1}\boldsymbol{p}$ for the velocity $\boldsymbol{v}$ wherever it appears. In practice the choice $\sigma = 1/2$ is used to avoid spurious damping (it can be demonstrated for a simple model problem); this certainly would appear to be desirable in the setting of molecular dynamics. For $\eta = 0$ we then arrive at the Verlet method. For other values of η the scheme is clearly implicit, which likely is the reason it is rarely used in molecular simulation, although it is popular in structural mechanics. The implicit Newmark methods are not symplectic, but a related family of symplectic methods can be constructed by replacing interpolated forces by forces evaluated at interpolated positions [395].

2.5.4 Multiderivative Methods

All of the methods discussed above rely on computing values of the vector field (i.e. momenta and forces) only. However, as we saw earlier in this chapter, in the setting of a Taylor series method, we may approximate a single step by

$$z_{n+1} = z_n + h\dot{z}_n + \frac{h^2}{2}\ddot{z}_n + \ldots + \frac{h^k}{k!}z_n^{(k)},$$

where it is possible to make use of higher order derivatives of the solution in formulating the method. Then using the differential equation, the time derivatives may be replaced by elementary differentials of the vector field. This same idea can be used in a more sophisticated way to improve the accuracy of molecular dynamics methods. Letting $H(\boldsymbol{q},\boldsymbol{p}) = \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2 + U(\boldsymbol{q})$, the Takahashi-Imada method [355] (also known as Rowlands' method [316]) has the same form as the Verlet method

$$\hat{\boldsymbol{P}} = \boldsymbol{p} - (h/2)\nabla\tilde{U}(\boldsymbol{q}),$$
$$\boldsymbol{Q} = \boldsymbol{q} + h\boldsymbol{M}^{-1}\hat{\boldsymbol{P}},$$
$$\boldsymbol{P} = \hat{\boldsymbol{P}} - (h/2)\nabla\tilde{U}(\boldsymbol{Q}),$$

where the corresponding potential energy function is